

SMART WASTE MANAGEMENT FOR SUSTAINABLE CAMPUSES



Smarter Campus. Cleaner Tomorrow.
Sustainable Future.



IoT Enabled
Smart Monitoring



Real-time
Analytics



Efficient
Operations



Engaged
Community



Reduce Waste
Minimize waste
at source



Smart Segregation
AI-powered sorting
for efficiency



Lower Emissions
Reduce carbon footprint
on campus



Sustainable
Campuses



Better
Tomorrow



Educate
& Empower



Together for
a Greener Future

The Challenge

Campus waste management today faces critical operational and environmental challenges that demand an intelligent solution.

🗑️ Overflowing Waste Bins

Bins reach capacity without timely collection, causing hygiene and health concerns across campus.

🕒 Delayed Waste Collection

Manual monitoring leads to slow response times and inefficient scheduling of collection activities.

🚚 Inefficient Collection Routes

Unoptimized routes result in unnecessary vehicle trips, higher fuel consumption, and increased emissions.

💰 Higher Operational Costs

Environmental impact from unnecessary vehicle trips drives up waste management expenditure significantly.

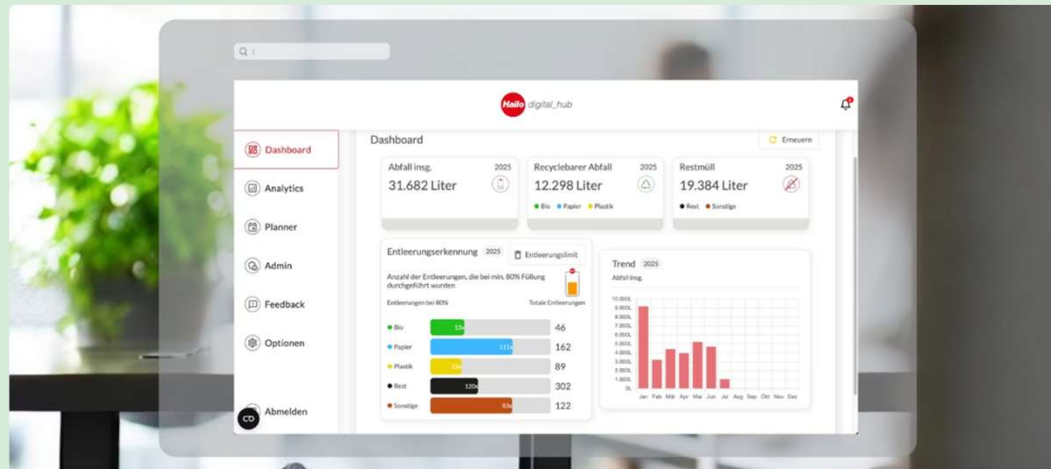
📌 **The Need:** A smart system that can monitor waste, generate alerts, and optimise collection activities in real time.



Our Solution — EcoSort

AI

EcoSort AI is an intelligent waste management platform designed to transform campus operations through data-driven automation and analytics.



- **Real-Time Bin Monitoring**
Continuously track fill levels across all 35 campus bins with live data updates.
- **Automated Collection Alerts**
Instant notifications for critical, warning, and info-level bin status changes.
- **Route Optimisation**
AI-generated efficient collection routes that reduce travel distance and fuel usage.
- **Sustainability Analytics Dashboard**
Visualise waste trends, recycling rates, and cost metrics in a single unified view.



Goal: Improve efficiency, reduce costs, and support sustainable campus operations.

How EcoSort AI Works

EcoSort AI follows a structured pipeline that transforms raw waste data into actionable insights for the campus waste management team.



Data flows seamlessly from historical records through monitoring, alerting, and optimisation stages before reaching the management team via the analytics dashboard.

Technologies Used

Python	Streamlit
Pandas	Plotly
Folium	SciPy



Core Functionalities



Smart Bin Monitoring

Track all **35 campus bins** with real-time fill level monitoring, ensuring no bin overflows unnoticed.



Automated Alerts

Three-tier alert system — **Critical, Warning, and Info** — prevents bin overflow and enables proactive response.



Route Optimisation

AI-calculated **efficient collection routes** significantly reduce fuel consumption and travel time for waste vehicles.



Sustainability Dashboard

Comprehensive visualisation of **waste trends, recycling rates**, and cost analysis for informed decision-making.

Impact & Benefits

EcoSort AI delivers measurable environmental and operational improvements for campus waste management.

25%

Reduction in Collection Trips

Fewer unnecessary vehicle trips across campus, lowering emissions.

15%

Increase in Recycling Efficiency

Improved sorting and recycling rates through data-driven insights.

20%

Reduction in Management Costs

Lower operational expenditure through optimised collection and routing.



Environmental Impact

- 25% reduction in collection trips
- 10–15% increase in recycling efficiency
- Lower carbon footprint from optimised routing



Operational Benefits

- 20% reduction in waste management costs
- Faster response to overflowing bins
- Data-driven decision making for management teams

Conclusion & Future Scope

Why EcoSort AI?

"Building Cleaner, Smarter, and More Sustainable Campuses through Data-Driven Waste Management."

These next-phase capabilities expand EcoSort AI from an intelligent campus solution into a scalable platform for cleaner, more efficient, and more sustainable waste management.



Smart



Sustainable



Scalable



cost
effective